

Hyperparameter Tuning of GDBT Models for Prediction of Heart Disease

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ABSTRACT

According to World Health Organization, heart disease has remained the leading cause of death for the last twenty years. Patients with heart disease will have palpitations, chest pain, dyspnea and other characteristics. How to diagnose heart disease based on these characteristics is an important research topic for doctors and scientists. Classifying heart disease with machine learning is known as a reliable and accurate heart disease detection technique. In this study, we aim to find a good model for predicting heart disease. To find the best model, we compare three GDBT (Gradient Boosting Decision Tree) models i.e., XGBoost, LightGBM and CatBoost to identify whether a person has heart disease or not. We compare the three models tuning the hyperparameters to improve heart disease predicting. To compare results, we create three base models with default hyperparameters. The findings demonstrate that the CatBoost with hyperparameter tuning model as the most appropriate heart disease prediction model with accuracy of 90%. Results indicated that tuning the hyperparameter improved modeling results.

Keywords-Heart disease; XGBoost; LightGBM; CatBoost; Hyperparameter tuning

1 INTRODUCTION

According to data, heart disease has become the number one killer of human health. Every year about 17.5 million people die of heart disease, accounting for 30% of all deaths worldwide. Heart disease seriously affects people's life and health. Patients with heart disease will have palpitations, chest pain, dyspnea and other characteristics. How to diagnose heart disease based on these characteristics is an important research topic for doctors and scientists. Classifying heart disease with machine learning is known as a reliable and accurate heart disease detection technique. In this study, we used three Gradient Boosting Decision Trees (XGBoost, LightGBM and CATBoost) to classify heart disease based on its characteristics. First, we preprocess the data, including repeated value processing, missing value processing and outlier processing. Our preliminary analysis of the data, including the correlation between features, the relationship between gender, age, blood pressure and other features and the disease. Second, we build three Gradient Boosting Decision Trees for classification prediction of heart disease. The three Gradient Boosting Decision Trees are XGBoost, LightGBM and CATBoost. Third, for each classifier, we conduct two types of experiments with default hyperparameters and adjustable hyperparameters, and compare their Accuracy, Recall, Roc_Auc and Precision metrics results. The experimental results show that using hyperparameter tuning will bring about a significant improvement in performance indicators.

The remainder of this paper is organized as follows. Related work in heart disease prediction is reviewed in Section II. In Section III, we describe the research methodology, data preprocessing method, hyperparameter tuning technique, three classifiers. Section IV compares the results of three classifier models with default parameters and adjustable hyperparameters. The last section concludes the paper and discusses future work.

2 RELATED WORKS

Lot of research has been carried out to predict heart disease using machine learning methods. Different implementation of machine learning methods such as Support Vector Machine, Logistic Regression, Bayesian Belief Network, etc. have been applied. Anirban Ghosh using five different classification models (Logistic Regression, Support Vector Machine, Random Forest Model, Naive Bayes classifier and Linear Discriminant analysis) to predict heart disease. The author thinks RFM is the most appropriate predictor of heart disease^[1].

Classifiers usually require a large number of data features to train. To identify the most important features that contribute to classifying the result, the author uses Sharpley value as an aid to analyze heart disease. To classify the heart disease data, the author uses random forest classifier with 20 decision trees^[2]. Pal et.al also used Random Forest model to predict heart disease. They obtained 86.9% accuracy with 90.6% sensitivity value and 82.7% specificity value^[3].

Polipireddy Srinivas and Rahul Katarya proposed a novel model hyOPTXg to identify cardiovascular disease^[4]. Their model utilized the XGBoost and OPTUNA technique. They also applied data preprocessing and hyper-parameter tuning techniques.

Mohammed Marouane Saim and Hassan Ammor compared three machine learning algorithms(SVM, Logistic Regression and KNN)to predict cardiovascular diseases^[5]. In their project the accuracy of Suppoert Vector Machine is 86.64%.

3 METHODOLOGY

In this study, we mainly adopt three classifiers, they are CatBoost, LightGBM and XGBoost. For each classifier, we adopt the following process: firstly prepare the data and perform data preprocessing. For the preprocessed data, we use two methods, one is to use the default hyperparameter for the classifier, and the other is to adjust the hyperparameter. Using the method of hyperparameter adjustment, the first step is to set the goal of hyperparameter learning and perform hyperparameter learning, and then classify the data with the learned hyperparameters. Finally, compare the performance metrics of the classifier using hyperparameter tuning and using the default hyperparameters. Figure 1. shows the workflow of hyperparameter optimization. This section includes the dataset description, data preprocessing techniques, classification algorithms and hyperparameter tuning techniques.

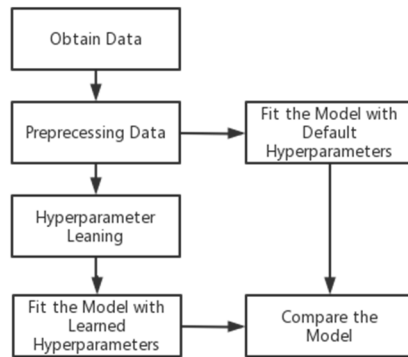


Figure1. Workflow of the methodology

3.1 The dataset

This dataset contains 12 columns, The HeartDisease is the output variable, while eleven input ones including age, sex, ChesPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExercideAngina, Oldpeak, ST_Slope are input variables. The database contains a total of 918 records with no missing values, so we dot need to deal with missing values. Percentage of HeartDisease is 55.34%, total of 508 customers, percentage of customer did not heartDisease is 44.66%, total of 410 customers. It can be seen from the proportion of HeartDisease that there is no imbalance in the data. The database contains 7 numerical columns and 5 categorical columns. Table 1. gives the statistics of numerical columns.

TABLE 1. THE STATISTICS OF NUMERICAL COLUMNS

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak
count	918	918	918	918	918	918
mean	53.5	132.3	198.7	0.2	136.8	0.8
std	9.4	18.5	109.3	0.4	25.4	1.0
min	28.0	0.0	0.0	0.0	60.0	-2.6
max	77.0	200.0	603.0	1.0	202.0	6.2

3.2 Data Preprocessing

In the data preprocessing stage, we map the 5 categorical variables into numeric types. As for ChestPainType, according to its statistical characteristics, we map ASY to 1, NAP to 2, ATA to 3, and TA to 4.

For outlier detection, we define a function to plot both the boxplot and distribution of numerical features. It can be seen from the figure that there are outlier in the three attributes of RestingBP, Cholesterol and Oldpeak. We delete the corresponding records. Figure 2. gives the boxplot and distribution of numerical features.

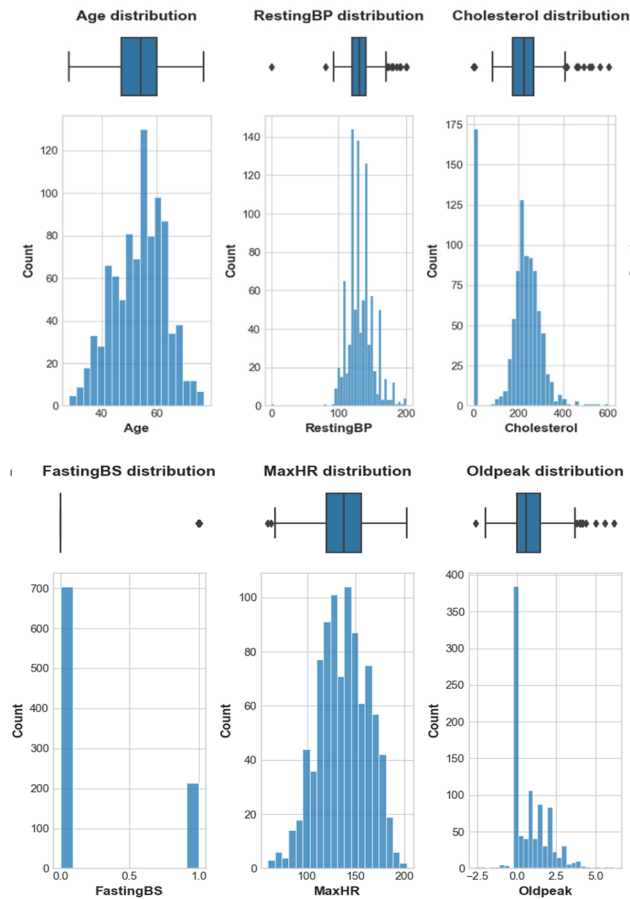


Figure 2. the boxplot and distribution of numerical features.

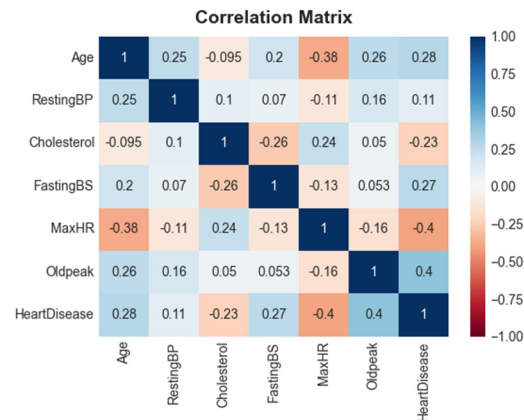


Figure 3. Correlation of numerical features

TABLE 2. HYPERPARAMETERS OBTAINED BY THE BAYESIAN OPTIMIZATION METHOD

Name	objective	Hyperparameter	Hyperparameter Importance
CatBoost	"objective": "Logloss","colsample_bylevel": 0.01, 0.1,"depth", 1, 12,"boosting_type": ["Ordered", "Plain"],"bootstrap_type": ["Bayesian", "Bernoulli", "MVS"]	colsample_bylevel: 0.09932226640716055, depth: 10, boosting_type: Plain, bootstrap_type: MVS	colsample_bylevel: 0.77,depth:0.12,boosting_type: 0.08, bootstrap_type: 0.03
LightGBM	"objective": "binary","metric": "binary_logloss","verbosity": -1, "boosting_type": "dart","num_leaves", 2, 2000,"max_depth", 3, 12,"lambda_l1", 1e-8, 10.0, log=True,"lambda_l2", 1e-8, 10.0, log=True,"num_leaves", 2, 256,"feature_fraction", 0.4, 1.0,"bagging_fraction", 0.4, 1.0,"bagging_freq", 1, 7,"min_child_samples", 5, 100	num leaves: 576, max_depth: 6, lambda_l1: 1.3159635619769443e- 06, lambda_l2: 0.06953260196279283, feature_fraction: 0.7930046206900255, bagging_fraction: 0.9432884010278475, bagging_freq: 2, min_child_samples: 22	min_child_samples: 0.72,feature_fraction:0.16, num_leaves: 0.06, bagging_fraction: 0.03,bagging_freq:0.02,max _depth: 0.01,lambda_l1: 0,lambda_l2:0
XGBoost	"verbosity": 0,"objective": "binary:logistic","tree_method": "exact","booster", ["gbtree", "gblinear", "dart"],"lambda", 1e-8, 1.0, log=True, "alpha", 1e-8, 1.0, og=True,"subsample", 0.2, 1.0,"colsample_bytree", 0.2, 1.0	booster: gblinear, lambda: 0.002146449268433026, alpha: 9.721458639113031e-05, subsample: 0.5699324732532789, colsample_bytree: 0.9414920603461235,	subsample: 0.62,colsample_bytree: 0.28,alpha: 0.05,booster:0.04,alpha:0.01

Figure 3 gives the correlation of numerical features. Based on the matrix, we can observe weak level correlation between the numerical features and the target variable. Oldpeak has a positive correlation with the heart disease. Maximum heart rate has negative correlation with the heart disease. Cholesterol has negative correlation with the heart disease.

3.3 Machine learning methods

In the field of machine learning, the Boosting algorithm is suitable for the field of structured data with few training samples. Commonly used Boosting algorithms are XGBoost, LightGBM and CatBoost.

XGBoost, also known as extreme gradient boosting tree, is an implementation of the boosting algorithm. XGBoost is a tool for massively parallel boosting trees. It is characterized by fast calculation speed and good model performance, which can be used in classification and regression problems. In the selection of the optimal split point, XGboost performs pre-sorting first, and then calculates the objective function gain of all samples split by these split points for all split points of all features. This process has great space and time complexity^[5].

LightGBM(Light Gradient Boosting Machine) is a framework for implementing the GBDT algorithm. It uses a histogram algorithm to convert traversal samples into traversal histograms, which greatly reduces the time complexity;It adopts the growth strategy based on the Leaf-wise algorithm to build the tree, which reduces a lot of unnecessary computation; It uses a histogram algorithm to convert stored eigenvalues into stored bin values, and does not require the index of eigenvalues to samples, reducing memory consumption; Therefore, LightGBM can use the parameter "maximum tree depth" to limit the depth of the tree and avoid overfitting^[6].

Catboost can automatically handle categorical features. Based on the statistics of categorical features, calculate the frequency of categorical features, add hyperparameters, and then generate new numerical features. Catboost makes it robust by reducing the need for many hyperparameter tuning and reducing the chance of overfitting. In addition, CatBoost also solves the problem of gradient bias (Gradient Bias) and prediction shift (Prediction shift), thereby reducing the occurrence of overfitting, thereby improving the accuracy and generalization ability of the algorithm^[7].

3.4 Hyperparameter tuning

Hyperparameter tuning techniques mainly include Grid Search, Random Search, Evolutionary Algorithms and Bayesian Optimization. In this paper, we use Bayesian Optimization methods^[8]. Table 2. gives the parameter objectives for

optimization and the hyperparameters obtained by the Bayesian Optimization method. The second column in the table is the learning objective, the third column is the learned hyperparameter values, the fourth column is the learned hyperparameter importance.

3.5 Result

In this paper, we use accuracy, recall, ROC-AUC, and precision as metrics to evaluate the performance of our methods.

Accuracy refers to the ratio of the number of correctly predicted samples to the total observations, and it does not consider whether the predicted samples are positive or negative.

$$Accuracy = (TP + TN) / (TP + TN + FN + FP) \quad (1)$$

Recall is the ratio of the number of correctly predicted positive samples to the all observations in actual class - yes.

$$Recall = TP / (TP + FN) \quad (2)$$

ROC-AUC computes the area under the ROC curve and is defined as the false-positive rate versus the true positive rate.

$$TPR = TP / (TP + FN) \quad (3)$$

$$FPR = FP / (FP + TN) \quad (4)$$

Precision is defined as the ratio of correctly predicted positive observations divided by the number of all predicted positive samples, that is, how many of all predicted positive samples are true positive samples

$$Precision = TP / (TP + FP) \quad (5)$$

Table 3. presents the values of Accuracy, Recall, Roc-Auc, and Precision for the three classifiers with and without hyperparameter tuning. As can be seen from the table, using hyperparameter tuning has a certain amount of improvement in Accuracy, Recall, Roc-Auc metrics. Three classifiers Catboost with optuna has the highest performance.

TABLE 3. CLASSIFICATION RESULTS

	Accuracy	Recall	Roc_Auc	Precision
Catboost	0.8950	0.8990	0.8953	0.8725
Catboost_optuna	0.9041	0.9192	0.9054	0.8750
LightGBM	0.8804	0.8902	0.8782	0.9068
LightGBM_optuna	0.8877	0.9024	0.8843	0.9080
XGBoost	0.8767	0.8788	0.8769	0.8529
XGBoost_optuna	0.8858	0.9091	0.8879	0.8491

We also compare our method with classification methods such as logistic regression, support vector machine, random forest model, naive bayes classifier and linear discriminant analysis used by Anirban Ghosh^[1]. The results show that using these three GDBT models has a more prominent performance on Accuracy, Recall, and Precision. Figure 4.gives performance comparison of different classifiers.

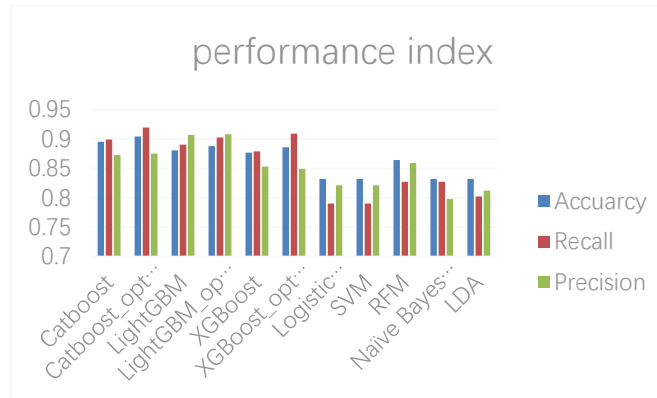


Figure 4. Performance comparison of different classifiers

4 CONCLUSION

In this study, in order to find a best model using GBDT models to predict heart disease we compare three GBDT models i.e., XGBoost, LightGBM and CatBoost to predict whether a person has heart disease or not. We compare the models tuning the hyperparameter to improve heart disease predicting. To compare results, we create three base models with default hyperparameter. The study found that using the CatBoost hyperparameter tuning method for heart disease prediction can achieve an average accuracy of 90%. Results also indicated that tuning the hyperparameters improved modeling results.

REFERENCES

- [1] Anirban Ghosh , Sushovon Jana. A Study on Heart Disease Prediction using Different Classification Models based on Cross Validation Method[J],International Journal of Engineering Research &Technology,2022,11(6):67-71.
- [2] D. Scapin, G. Cisotto, E. Gindullina, and L. Badia, Shapley value as an aid to biomedical machine learning: a heart disease dataset analysis[C],2022 22nd IEEE International Symposium on Cluster, Cloud and Internet Computing (CCGrid), Italy, May,2022: 933-939.
- [3] N. M. Lutimath, N. Sharma and B. K. Byregowda.Prediction of Heart Diseases using Random Forest[C],2021 Emerging Trends in Industry 4.0 (ETI 4.0), 2021:1-4.
- [4] Polipireddy Srinivas, Rahul Katarya,hyOPTXg: OPTUNA hyper-parameter optimization framework for predicting cardiovascular disease using XGBoost[J],Biomedical Signal Processing and Control,2022,73(3):103456-1034.
- [5] Saim, Mohammed & Hassan, Ammor. Comparative study of machine learning algorithms (SVM, Logistic Regression and KNN) to predict cardiovascular diseases[C]. E3S Web of Conferences. 2022, 35
- [6] Chen T, Guestrin C. Xgboost: A scalable tree boosting system[C], Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining. ACM, 2016: 785-794.
- [7] Ke G, Meng Q, Finley T, et al. Lightgbm: A highly efficient gradient boosting decision tree[C], Advances in Neural Information Processing Systems. 2017: 3146-3154.
- [8] Prokhorenkova L, Gusev G, Vorobev A, et al. CatBoost: unbiased boosting with categorical features[C], Advances in Neural Information Processing Systems. 2018: 6638-6648.
- [9] T. Akiba, S. Sano, T. Yanase, T. Ohta, M. Koyama, Optuna: A next-generation hyperparameter optimization framework[C], Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min. 2019: 2623–2631.